Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

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List of PAPILLON Investigators

Principal investigator	Clinical site
Andres Aguilar	HOSP. UNIV. QUIRON DEXEUS
Jorge Arturo Alatorre	Health Pharma Professional Research
Adlinda Alip	University Malaya Medical Centre
Andrea Ardizzoni	A.O.U Sant'Orsola-Malpighi
Oscar Gerardo Arrieta	Instituto Nacional de Cancerologia
Isabel Azevedo	Instituto Portugues de Oncologia
Koichi Azuma	Kurume University Hospital
Clarissa Baldotto	Instituto D'Or de Pesquisa e Ensino (IDOR)
Ana Barroso	Centro Hospitalar de Vila Nova de Gaia/Espinho, E.P.E.
Ullas Batra	Rajiv Gandhi Cancer Institute & Research Centre
Reyes Bernabe Caro	HOSP. VIRGEN DEL ROCIO
Ahmet Bilici	Medipol Mega University Hospital
Bivas Biswas	Tata Medical Center
Samantha Bowyer	Sir Charles Gairdner Hospital
Michael Boyer	Chris O'Brien Lifehouse
Farastuk Bozorgmehr	Thoraxklinik am Universitätsklinikum Heidelberg
Emilio Bria	IRCCS GEMELLI
Enric Carcereny Costa	INST. CAT. D'ONCOLOGIA-BADALONA
Gee-chen Chang	Chung Shan Medical University Hospital
Yuan Chen	TongJi Hospital of TongJi Medical College of Huazhong University of Science & Technology
Jianhua Chen	Hunan Cancer Hospital
Susanna Cheng	Sunnybrook Health Sciences Centre
Byoung Chul Cho	Severance Hospital, Yonsei University Health System
Irfan Cicin	Trakya University Medical Faculty
Timucin Cil	Adana City Hospital
Alexis Cortot	CHR Hôpital Calmette
Sulene Cunha Souza Oliveira	Liga Norte Riograndense Contra O Cancer
Pongwut Danchaivijitr	Siriraj Hospital
Javier De Castro De Castro	HOSP. UNIV. LA PAZ
Gilberto De Castro Junior	Fundacao Faculdade de Medicina – Instituto do Cancer do Estado de Sao Paulo
Angelo Delmonte	Istituto Scientifico Romagnolo per lo Studio e la Cura dei Tumori
Christophe Dooms	UZ Leuven
Michaël Duruisseaux	Hospices Civils de Lyon HCL
Rafal Dziadziuszko	Uniwersyteckie Centrum Kliniczne
Yong Fang	Sir Run Run Shaw Hospital, Zhejiang University School of Medicine
Enriqueta Felip	HOSP. UNIV. VALL D HEBRON
Miguel Fernández Sanmamed	CLINICA UNIV. DE NAVARRA
Daichi Fujimoto	Wakayama Medical University Hospital

Maria Del Rosario Garcia Campelo	HOSP. UNIV. A CORUÑA
Pilar Garrido	Hosp. Univ. Ramon Y Cajal
Ana Caroline Gelatti	União Brasileira de Educação e Assistência – Hospital São Lucas da PUCRS
Nicolas Girard	Institut Curie
Jonathan Goldman	University of California Los Angeles
Jerome Goldschmidt	Blue Ridge Cancer Care
Frank Griesinger	Pius-Hospital Oldenburg
Mahmut Gumus	Goztepe Prof. Dr. Suleyman Yalcin Sehir Hastanesi
Zhongliang Guo	Shanghai East Hospital
Xiaosheng Hang	Hospital of Jiangnan University
Osamu Hataji	Matsusaka Municipal Hospital
Hidetoshi Hayashi	Kindai University Hospital
Yi Hu	Chinese PLA General Hospital
Jen-yu Hung	Kaohsiung Medical University Chung-Ho Memorial Hospital
Dolores Isla	Hosp. Clinico Univ. Lozano Blesa
Minish Jain	Noble Hospital Pvt Ltd
Tetsuji Kawamura	National Hospital Organization Himeji Medical Center
Stephan Disean Kendall	Utah Cancer Specialists
Chul Kim	Georgetown University Hospital
Yu Jung Kim	Seoul National University Bundang Hospital
Sang-we Kim	Asan Medical Center
Nikolay Kislov	Yaroslavl Regional Clinical Oncology Hospital
Satoru Kitazono	The Cancer Institute Hospital of JFCR
Dariusz Kowalski	Narodowy Instytut Onkologii im. Marii Sklodowskiej-Curie – Panstwowy Instytut Badawczy
Kang-yun Lee	Taipei Medical University Shuang Ho Hospital
Se-hoon Lee	Samsung Medical Center
Young Joo Lee	National Cancer Center
Yun-gyoo Lee	Kangbuk Samsung Hospital
Wen Li	The Second Affiliated Hospital of Zhejiang University College of Medicine
Chun Sen Lim	Hospital Sultan Ismail
Zhe Liu	Beijing Chest Hospital, Capital Medical University
Caigang Liu	Shengjing Hospital of China Medical University
Baogang Liu	Harbin Medical University Cancer Hospital
Shun Lu	Shanghai Chest Hospital
Dongqing Lv	Taizhou Hospital of Zhejiang Province
M.V.T Krishna Mohan	Basavatarakam Indo-American Hospital
Margarita Majem	HOSP. DE LA SANTA CREU I SANT PAU
Shingo Matsumoto	National Cancer Center Hospital East
Ofer Merimsky	Tel-Aviv Sourasky Medical Center
Josiane Mourao Dias	Fundacao Pio XII
Rajnish Nagarkar	HCG Manavta Cancer Centre

Misako Nagasaka	University of California Irvine
Hiromi Nagashima	Iwate Medical University Hospital
Hovav Nechushtan	Hadassah Medical Center
Vanita Noronha	Tata Memorial Hospital
Sebahat Ocak	CHU UCL Namur – Site Godinne
Francesca Ogliari	IRCCS Ospedale San Raffaele
Masahide Oki	National Hospital Organization Nagoya Medical Center
Akira Ono	Shizuoka Cancer Center
Gyula Ostoros	Orszagos Koranyi Tbc es Pulmonologiai Intezet
Scott Owen	McGill University Health Centre
Ozgur Ozyilkan	Başkent University Medical Faculty Adana Application and Research Center
Luis Paz-Ares Rodriguez	HOSP. UNIV. 12 DE OCTUBRE
Nir Peled	Shaare Zedek Medical Center
Dmitry Ponomarenko	Irkutsk Regional Oncology Dispensary
Sanjay Popat	The Royal Marsden NHS Trust
Steven Powell	Sanford Health
Naiyarat Prasongsook	Phramongkutklao Hospital and Medical College
Claudia Proto	Fondazione IRCCS Istituto Nazionale dei Tumori
Inmaculada Ramos Garcia	HOSP. VIRGEN DE LA VICTORIA
Gary Richardson	Cabrini Medical Centre
Joshua Sabari	Langone Health at NYC University, NYU School of Medicine
Victor Santos	Ministerio da Saude – Instituto Nacional do Cancer
Yuki Sato	Kobe City Medical Center General Hospital
Mehmet Ali Nahit Sendur	Ankara Bilkent City Hospital
Hong Shen	The Second Affiliated Hospital of Zhejiang University College of Medicine
Catherine Shu	Columbia University Medical Center
Felipe José Silva Melo Cruz	Núcleo de Pesquisa São Camilo
Hwoei Fen Soo Hoo	Hospital Pulau Pinang
Hector Jose Soto Parra	Azienda Ospedaliero Univ. Policlinico Gaspare Rodolico
Alexander Spira	Virginia Cancer Specialists
David Stewart	The Ottawa Hospital Cancer Centre
Daniil Stroyakovskiy	Moscow City Oncology Hospital № 62
Ping Sun	Yantai Yuhuangding Hospital
Hiroshi Tanaka	Niigata Cancer Center Hospital
Ke-Jing Tang	The First Affiliated Hospital, Sun Yat-sen University
Encarnacao Teixeira	HOSP. CUF DESCOBERTAS
Muthukkumaran Thiagarajan	Hospital Kuala Lumpur
Rémi Veillon	CHU Bordeaux
Huijie Wang	Fudan University Shanghai Cancer Center
Donglin Wang	Chongqing University Cancer Hospital
Naohiro Watanabe	Aichi Cancer Center Hospital
Lin Wu	Hunan Cancer hospital

Yong Xu	West China Hospital, Sichuan University
Yu Yao	The First Affiliated Hospital of Xi'an Jiaotong University
Ozan Yazici	Gazi University Hospital
Toshihide Yokoyama	Kurashiki Central Hospital
Hiroshige Yoshioka	Kansai Medical University Hospital
Yanqiu Zhao	Henan Cancer Hospital
Jun Zhao	Beijing Cancer Hospital
Caicun Zhou	Shanghai Pulmonary Hospital
Jianying Zhou	The First Affiliated Hospital, Zhejiang University College of Medicine
Bogdan Zurawski	Centrum Onkologii im. Prof. F. Lukaszczyka

SUPPLEMENTARY METHODS

Additional Inclusion and Exclusion Criteria

Documented primary epidermal growth factor receptor (*EGFR*) exon 20 insertion (Ex20ins)–activating mutation must have been performed by a Clinical Laboratory Improvement Amendments–certified (sites in the USA) or an accredited (sites outside of the USA) local laboratory. A copy of the mutation analysis must have been submitted during screening.

Patients must have at least 1 measurable lesion according to Response Evaluation Criteria in Solid Tumours (RECIST), version 1.1. If only 1 measurable lesion exists, it may be used for the screening biopsy (if required for submission of tumor tissue) if the baseline tumor assessment scans are performed at least 14 days after the biopsy. Localized radiotherapy to the lung must be completed at least 6 months prior to randomization. Palliative radiation to other sites must be completed at least 7 days prior to randomization. Patients must have an Eastern Cooperative Oncology Group performance status of 0 or 1.

Patients must not have received any systemic treatment at any time for locally advanced stage III or metastatic stage IV disease, with the following exceptions: adjuvant or neoadjuvant platinum-based chemotherapy was allowed if completed at least 12 months prior to signing the informed consent form. Prior monotherapy with an approved EGFR tyrosine kinase inhibitor (TKI; ie, gefitinib, erlotinib, afatinib, dacomitinib, or osimertinib) as nonstandard first-line therapy for the treatment of locally advanced or metastatic disease was allowed if: 1) the treatment duration did not exceed 8 weeks, 2) lack of disease response was documented (radiographically) by an increase in tumor burden (a copy of the computerized tomography report showing the increase in tumor burden from baseline should be submitted), 3) associated toxicities have resolved to baseline, and 4) the EGFR TKI was discontinued at least 2 weeks or 4 half-lives prior to randomization, whichever was longer. Prior therapy with investigational EGFR TKI agents that target Ex20ins mutations, including TAK788 or poziotinib, was not allowed.

Patients must not have untreated brain metastases. Patients with asymptomatic or previously treated, stable brain metastases were eligible provided they were off of corticosteroid treatment for at least 2 weeks prior to randomization.

Patients were not eligible if they had received an investigational medication within 6 weeks before randomization. Patients must not have a concurrent or prior malignancy other than the disease under study or have an uncontrolled illness.

Management of Common Adverse Events

Recommendations for proactive and reactive management of rash and infusion-related reactions were incorporated in the protocol. Rash was typically managed by topical steroids and oral antibiotics. Oral steroids with or without systemic antibiotics could be added based on severity of the rash. Infusion related reactions were typically managed by an infusion interruption, symptomatic treatment, and re-initiation upon resolution of an infusion reaction. Management of chemotherapy-related, including gastrointestinal, toxicities was per local guidance and labeling. All management details are available in the protocol.

Statistical Analysis

Objective response was analyzed using a stratified logistic regression model stratified by ECOG performance status (0 or 1) and history of brain metastases (yes or no). Overall survival was analyzed using the similar methodology and model as for the primary analysis of progression-free survival.

A hierarchical testing approach was used. Therefore, an objective response rate and overall survival analysis, each with an overall 2-sided alpha of 0.05, will occur if progression-free survival shows statistical significance. The test of objective response rate will be conducted before the test for overall survival.

The analysis of overall survival will be performed at 2 timepoints. The first will occur at the time of primary analysis of progression-free survival when approximately 85 deaths overall are anticipated. Based on the O'Brien Fleming alpha spending approach, a 2-sided alpha of 0.0008 will be allocated to the interim analysis. The final overall analysis will be at approximately 48 months after the first patient is randomized, when approximately 210 deaths overall are anticipated . The final analysis will be conducted at a 2-sided alpha of 0.0498.

Crossover After Disease Progression to Second-line Amivantamab Monotherapy

Patients randomized to receive chemotherapy may crossover to amivantamab monotherapy after disease progression once confirmed by blinded independent central review. Additional inclusion/exclusion criteria for the crossover arm include: 1) no anticancer or investigational

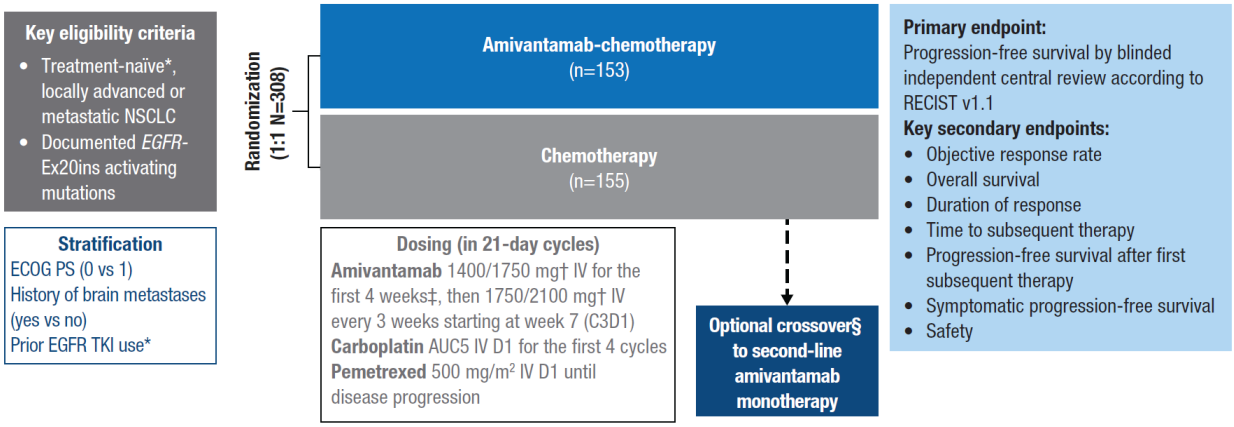
therapy following discontinuation of chemotherapy and 2) all toxicities must have resolved to grade ≤1 severity (except for alopecia, which may be grade 2).

Trial assessments followed the same schedule as the randomized portion except relative to initiation of second-line amivantamab monotherapy. Endpoints included objective response rate, duration of response, and safety. All analyses were descriptive.

SUPPLEMENTARY RESULTS

Supplementary Figures

Figure S1. PAPILLON Study Design.



AUC, area under the curve; C, cycle; D, day; ECOG PS, Eastern Cooperative Oncology Group performance status; EGFR, epidermal growth factor receptor; Exon20ins, exon 20 insertion mutation; IV, intravenous; NSCLC, non-small cell lung cancer; RECIST, Response Evaluation Criteria in Solid Tumours; TKI, tyrosine kinase inhibitor.

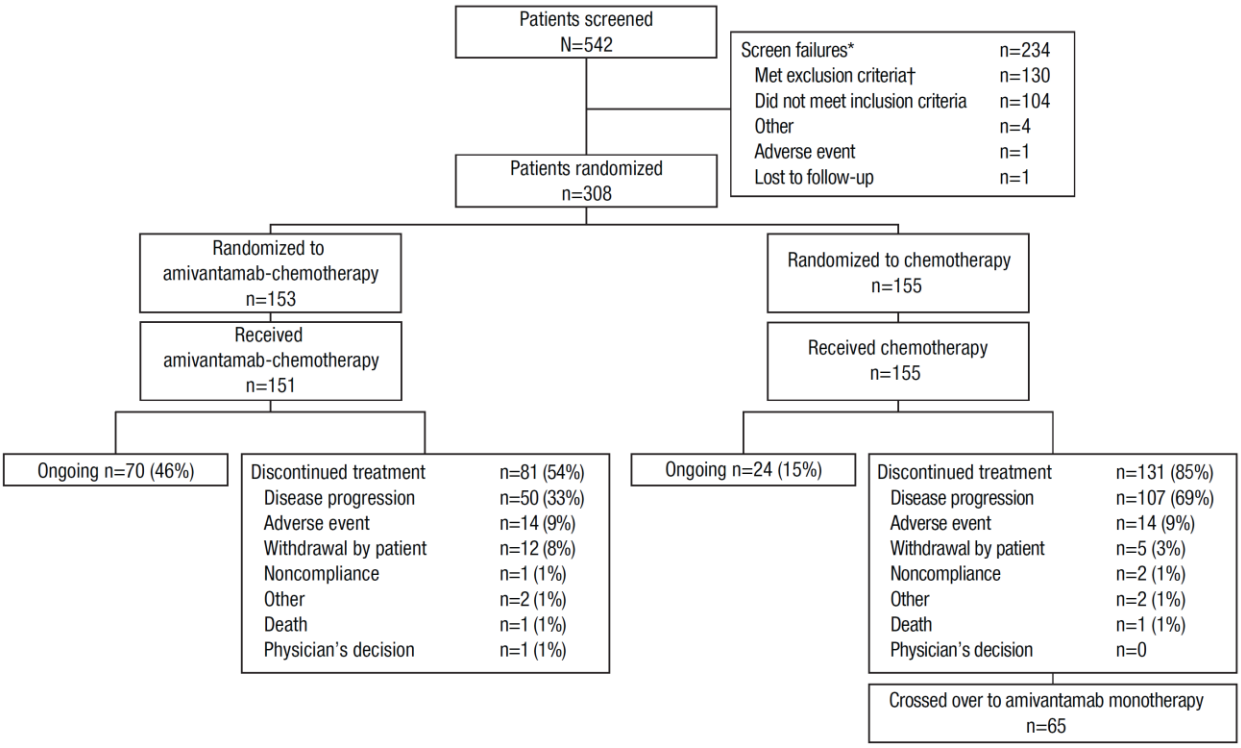
*Prior EGFR TKI use (transient monotherapy with common EGFR TKIs was allowed if lack of response was documented) was removed as a stratification factor for the primary analysis as only 4 patients received prior EGFR-TKI monotherapy.

†1750/2100 mg doses for patients of body weight 80 kg or greater.

‡Cycle 1, day 1/2 (split dose), 8, 15; cycle 2, day 1.

§Crossover is only allowed after confirmation of disease progression by blinded independent central review; dosing for second-line amivantamab monotherapy is the same as the main study.

Figure S2. Consolidated Standards of Reporting Trials (CONSORT) Diagram of Patient Disposition.

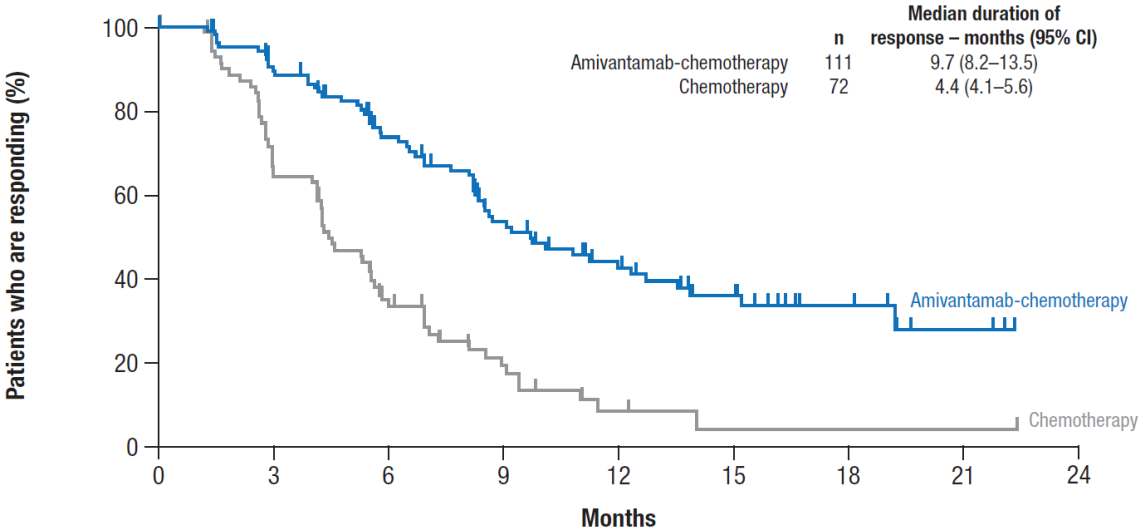


*Patients may be counted in more than 1 category.

†The most common reason for screen failure was untreated brain metastasis (n=56 patients).

Figure S3. Duration of Response.*

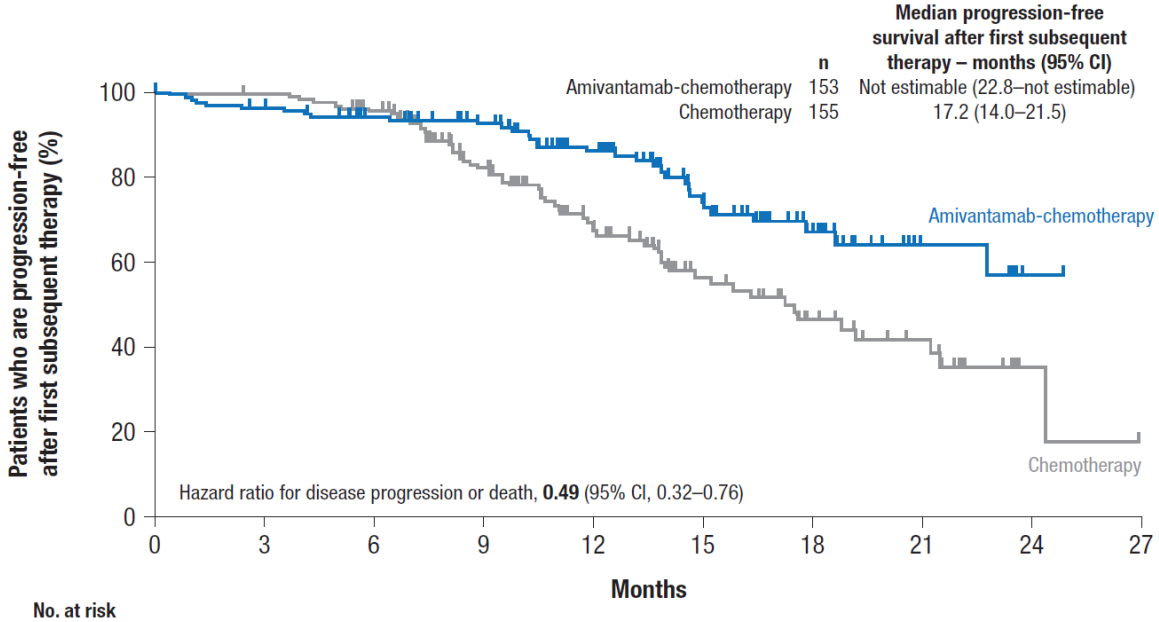
Shown is a Kaplan-Meier estimate of duration of response assessed by blinded independent central review. The efficacy analysis set included all randomized patients. The number of patients with measurable disease at baseline for each arm was 152 as determined by blinded independent central view. Tick marks indicate censoring of data. 95% CI widths have not been adjusted for multiplicity and cannot be used to infer definitive treatment effects. CI denotes confidence interval and SoD sum of diameters.



No. at risk									
Amivantamab-chemotherapy	111	91	66	42	28	18	8	3	0
Chemotherapy	72	45	23	10	3	1	1	1	0

Figure S4. Progression-free Survival After First Subsequent Therapy.

Shown is a Kaplan-Meier estimate of progression-free survival after first subsequent therapy. The efficacy analysis set included all randomized patients. Tick marks indicate censoring of data. 95% CI widths have not been adjusted for multiplicity and cannot be used to infer definitive treatment effects. CI denotes confidence interval and SoD sum of diameters.



No. at risk										
Amivantamab-chemotherapy	153	143	128	110	85	52	27	9	1	0
Chemotherapy	155	152	139	102	68	38	22	14	2	0

Supplementary Tables

Table S1. Representativeness of Study Participants

Category	**Details**
Disease under investigation	*EGFR*-Ex20ins NSCLC
Special considerations related to:	
Sex	*EGFR*-Ex20ins NSCLC affects more women than men (62% vs 38%, respectively) and also when compared with wild-type *EGFR* disease (*P*=0.003).[1,2]
Age	The median age of patients with *EGFR*-Ex20ins NSCLC has been estimated to be 60* to 63 years† depending on the study.[1,3]
Race or ethnic group	The prevalence of *EGFR*- Ex20ins NSCLC is higher in Asian patients versus wild-type *EGFR* (*P*=0.02).[3]
Geography	The frequency of *EGFR*-Ex20ins NSCLC is highest in Asia (0.1–4% of all NSCLCs and 1–5% of *EGFR*-mutated NSCLC) followed by the USA (0.5–2.6% of all NSCLCs and 5–12% of *EGFR*-mutated NSCLC).[2]
Other considerations	Compared with wild-type disease, *EGFR*-Ex20ins NSCLC is more commonly observed in patients who are never smokers (*P*<0.001).[3]
Overall representativeness of this trial	The demographic and epidemiologic characteristics of the PAPILLON patient population were generally consistent with available data on patients with *EGFR*-Ex20ins NSCLC.[1-3] There were more females than males, most patients did not have a smoking history, and the majority of patients were Asian‡.

EGFR, epidermal growth factor receptor; Ex20ins, exon 20 insertion mutation; NSCLC, non-small cell lung cancer.

*From a single-site analysis of 27 patients with EGFR-Ex20ins identified among 1086 patients with conclusive EGFR genotyping from 2004 to 2012.

†From a comprehensive genomic profile study of 263 patients with EGFR-Ex20ins identified among 14,483 NSCLC specimens.

‡Consistent with the EGFR-Ex20ins epidemiology, most of the patients in this study were Asian, followed by patients who are white, and with low numbers of patients of other races (ie, Black or African-American, American-Indian, or Alaska-Native, Multiple).

Table S2. Number of Treatment Cycles for the Individual Study Treatments

	Amivantamab-chemotherapy (n=151)			Chemotherapy (n=155)	
	Amivantamab*	**Carboplatin**	**Pemetrexed**	**Carboplatin**	**Pemetrexed**
No. of treatment cycles					
Mean (SD)	14 (7.9)	3.8 (0.7)	13.5 (7.9)	3.9 (0.5)	10.8 (5.8)
Median (range)	14 (1–39)	4 (1–4)	13 (1–34)	4 (1–5)	10 (1–37)
No. (%) receiving					
1–4 cycle(s)	16 (11)	151 (100)	20 (13)	154 (99)	17 (11)
5–8 cycles	21 (14)	0	24 (16)	1 (1)	39 (25)
9–12 cycles	32 (21)	0	31 (21)	0	49 (32)
13–16 cycles	36 (24)	0	30 (20)	0	31 (20)
≥17 cycles	46 (30)	0	46 (30)	0	19 (12)
Duration of treatment — months					
Mean (SD)	10.2 (6.1)	2.1 (0.6)	9.6 (6.0)	2.2 (0.5)	7.3 (4.1)
Median (range)	9.3 (0–26.9)	2.1 (0–3.8)	8.3 (0–24.9)	2.1 (0–3.9)	6.7 (0–25.3)

SD, standard deviation.

*Does not include patients in the chemotherapy arm who crossed over to second-line amivantamab monotherapy.

Table S3. First Subsequent Systemic Therapies, Including In-study Crossover

Therapy — no. of patients (%)	**Amivantamab-chemotherapy** (n=153)	**Chemotherapy** (n=155)
Any first subsequent therapy	43 (28)	94 (61)
EGFR-targeted or TKI-based regimen	12 (8)	82 (53)
Amivantamab	1 (1)	71 (46)*
EGFR TKI monotherapy	9 (6)	7 (5)
Alflutinib	4 (3)	2 (1)
Mobocertinib	2 (1)	4 (3)
Afatinib	1 (1)	1 (1)
Osimertinib	2 (1)	0
EGFR TKI combination	2 (1)	4 (3)
Afatinib + bevacizumab + docetaxel	0	1 (1)
Bevacizumab + osimertinib	1 (1)	0
Carboplatin + catequentinib + gemcitabine	0	1 (1)
Catequentinib + paclitaxel + sintilimab	0	1 (1)
Docetaxel + nintedanib	1 (1)	0
Other EGFR TKI	0	1 (1)
Chemotherapy or immunotherapy-based regimens	23 (15)	11 (7)
Chemotherapy alone	13 (8)	2 (1)
Pemetrexed	6 (4)	0
Docetaxel	3 (2)	0
Carboplatin + paclitaxel	2 (1)	0
Carboplatin + pemetrexed	0	1 (1)
Cisplatin + gemcitabine	1 (1)	0
Paclitaxel	1 (1)	0
Vinorelbine	0	1 (1)
Chemotherapy plus VEGF inhibitors	6 (4)	3 (2)
Docetaxel + ramucirumab	3 (2)	1 (1)
Bevacizumab + paclitaxel	3 (2)	0
Bevacizumab + carboplatin + paclitaxel	0	2 (1)
Chemotherapy combined with VEGF inhibitors and immunotherapy	1 (1)	3 (2)
Atezolizumab + bevacizumab + carboplatin + paclitaxel	1 (1)	1 (1)
Bevacizumab + paclitaxel + sintilimab	0	1 (1)

Bevacizumab + pemetrexed + sintilimab	0	1 (1)
Chemotherapy plus immunotherapy	2 (1)	1 (1)
Carboplatin + pembrolizumab + pemetrexed	1 (1)	0
Docetaxel + nivolumab	1 (1)	0
Docetaxel + tislelizumab	0	1 (1)
Immunotherapy only	1 (1)	2 (1)
Atezolizumab	1 (1)	0
PD-1/PD-L1 inhibitors	0	1 (1)
Pembrolizumab	0	1 (1)
Antineoplastic agents†	8 (5)	1 (1)

EGFR, epidermal growth factor receptor; PD-1, programmed cell death protein 1; PD-L1, programmed death-ligand 1; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor.

*A total of 65 patients in the chemotherapy arm crossed over to receive amivantamab monotherapy.

†Some may be investigational.

Table S4. Efficacy Endpoints (In-study Crossover)

Endpoint	Second-line amivantamab monotherapy* (n=65)
Objective response rate† — % of patients (95% CI)	39 (26–53)
Duration of response†	
No. of months — median (95% CI)	11.1 (5.2–not estimable)
% of patients with continued response at 6 months (95% CI)	62 (35–80)
% of patients with continued response at 12 months (95% CI)	40 (13–66)
Progression-free survival†	
No. of months — median (95% CI)	6.8 (4.4–9.6)
% of patients progression free at 6 months (95% CI)	52 (37–65)
% of patients progression free at 12 months (95% CI)	25 (12–41)
Overall survival	
No. of months — median (95% CI)	17.7 (12.1–not estimable)
% of patients alive at 6 months (95% CI)	91 (80–96)
% of patients alive at 12 months (95% CI)	70 (54–82)

CI, confidence interval.

*Patients in the chemotherapy arm who crossed over to receive amivantamab monotherapy (eligibility included disease progression verified by blinded independent central review); as of the time of data cutoff, the median follow-up was 9.8 months, the median treatment duration was 4.9 months, and the median number of treatment cycles was 8.

†Responses in the crossover arm were assessed by the investigator per Response Evaluation Criteria in Solid Tumours (RECIST), version 1.1.

Table S5. Adverse Events (In-study Crossover)

Event — no. (%)	Second-line amivantamab monotherapy* (n=65)	
Treatment emergent		
Any event	64 (98)	
Grade ≥3	24 (37)	
Any serious event	17 (26)	
Any event resulting in death	2 (3)	
Any event leading to		
Interruptions	22 (34)	
Reductions	8 (12)	
Discontinuations	7 (11)	
Adverse events†	All grades	Grade ≥3
Rash	39 (60)	3 (5)
Infusion-related reaction	32 (49)	1 (2)
Paronychia	32 (49)	2 (3)
Hypoalbuminemia	25 (38)	4 (6)
Aspartate aminotransferase increased	17 (26)	0
Alanine aminotransferase increased	15 (23)	0
Dermatitis acneiform	13 (20)	3 (5)
Constipation	12 (18)	0
Decreased appetite	12 (18)	0
Peripheral edema	12 (18)	0
COVID-19	11 (17)	0
Treatment related		
Any event	64 (98)	
Grade ≥3	16 (25)	
Any serious event	8 (12)	
Any event resulting in death	0	
Any event leading to		
Interruptions	14 (22)	
Reductions	8 (12)	
Discontinuations	4 (6)	
Adverse events†	All grades	Grade ≥3
Rash	39 (60)	3 (5)
Paronychia	32 (49)	2 (3)
Infusion-related reaction	31 (48)	1 (2)

Hypoalbuminemia	22 (34)	3 (5)
Aspartate aminotransferase increased	13 (20)	0
Dermatitis acneiform	13 (20)	3 (5)
Alanine aminotransferase increased	11 (17)	0
Peripheral edema	11 (17)	0

*Patients in the chemotherapy arm who crossed over to receive amivantamab monotherapy (eligibility included disease progression verified by blinded independent central review); as of the time of data cutoff, median follow-up was 9.8 months, the median treatment duration was 4.9 months, and the median number of treatment cycles was 8.

†Listed are adverse events that were reported in at least 15% of patients in any group.

Table S6. Efficacy Endpoints Assessed by Blinded Independent Central Review and the Investigator

	Blinded independent central review assessed		Investigator assessed*	
Endpoint	**Amivantamab-chemotherapy** (n=153)	**Chemotherapy** (n=155)	**Amivantamab-chemotherapy** (n=153)	**Chemotherapy** (n=155)
Progression-free survival				
No. of months — median (95% CI)	11.4 (9.8–13.7)	6.7 (5.6–7.3)	12.9 (11.4–16.7)†	6.9 (6.2–8.3)
Objective response rate‡ — % of patients (95% CI)				
All responders	73 (65–80)	47 (39–56)	66 (58–74)‖	43 (35–51)
Confirmed responders	67 (59–75)¶	36 (29–44)	62 (54–70)**	39 (31–47)
Best overall response‡ — no. (%)				
Complete response§	6 (4)	1 (1)	1 (1)	1 (1)
Partial response§	105 (69)	71 (47)	100 (65)	65 (42)
Stable disease	29 (19)	62 (41)	39 (25)	73 (47)
Progressive disease	4 (3)	16 (11)	5 (3)	14 (9)
Not evaluable	8 (5)	2 (1)	8 (5)	2 (1)
Duration of response‡				
No. of months — median (95% CI)* among all responders	9.7 (8.3–13.5)	4.4 (4.1–5.6)	13.5 (10.1–19.2)	6.8 (5.5–7.8)
No. of months — median (95% CI)* among confirmed responders	10.1 (8.5–13.9)	5.6 (4.4–6.9)	15.3 (10.9–not estimable)	6.9 (5.6–8.1)

CI, confidence interval.

The efficacy analysis set included all randomized patients.

*95% CI widths have not been adjusted for multiplicity and cannot be used to infer definitive treatment effects.

†Hazard ratio for disease progression or death was 0.38 (95% CI, 0.29 to 0.51).

‡No. of patients with measurable disease for each arm was 152 for assessment by blinded independent central review.

§Includes all responders.

‖Risk ratio was 1.44 (95% CI, 1.26 to 1.63).

¶Risk ratio was 1.55 (95% CI, 1.39 to 1.72).

**Risk ratio was 1.44 (95% CI, 1.26 to 1.62).

Table S7. Treatment-related Adverse Events

Event — no. (%)	**Amivantamab-chemotherapy** (n=151)				**Chemotherapy** (n=155)			
Any event	151 (100)				146 (94)			
Grade ≥3	100 (66)				57 (37)			
Any serious event	36 (24)				16 (10)			
Any event resulting in death	3 (2)				2 (1)			
Adverse events*	All	Grade 1	Grade 2	Grade ≥3	All	Grade 1	Grade 2	Grade ≥3
Neutropenia	85 (56)	8 (5)	28 (19)	49 (32)	68 (44)	8 (5)	26 (17)	34 (22)
Paronychia	84 (56)	34 (23)	40 (26)	10 (7)	0	0	0	0
Rash	81 (54)	40 (26)	24 (16)	17 (11)	10 (6)	8 (5)	2 (1)	0
Anemia	72 (48)	33 (22)	25 (17)	14 (9)	81 (52)	24 (15)	40 (26)	17 (11)
Infusion-related reaction	63 (42)	15 (10)	46 (30)	2 (1)	1 (1)	1 (1)	0	0
Leukopenia	55 (36)	12 (8)	26 (17)	17 (11)	47 (30)	13 (8)	29 (19)	5 (3)
Hypoalbuminemia	53 (35)	6 (4)	42 (28)	5 (3)	10 (6)	8 (5)	2 (1)	0
Thrombocytopenia	52 (34)	25 (17)	13 (9)	14 (9)	41 (26)	14 (9)	13 (8)	14 (9)
Nausea	50 (33)	37 (25)	13 (9)	0	63 (41)	46 (30)	17 (11)	0
Dermatitis acneiform	47 (31)	23 (15)	18 (12)	6 (4)	4 (3)	3 (2)	1 (1)	0
Decreased appetite	45 (30)	26 (17)	15 (10)	4 (3)	33 (21)	27 (17)	4 (3)	2 (1)
Alanine aminotransferase increased	41 (27)	29 (19)	7 (5)	5 (3)	50 (32)	33 (21)	15 (10)	2 (1)
Aspartate aminotransferase increased	39 (26)	35 (23)	3 (2)	1 (1)	47 (30)	43 (28)	3 (2)	1 (1)
Peripheral edema	39 (26)	26 (17)	12 (8)	1 (1)	9 (6)	7 (5)	2 (1)	0
Constipation	38 (25)	31 (21)	7 (5)	0	30 (19)	26 (17)	4 (3)	0
Stomatitis	36 (24)	16 (11)	18 (12)	2 (1)	7 (5)	6 (4)	1 (1)	0
Asthenia	27 (18)	11 (7)	10 (7)	6 (4)	22 (14)	15 (10)	3 (2)	4 (3)

Vomiting	27 (18)	15 (10)	9 (6)	3 (2)	23 (15)	17 (11)	5 (3)	1 (1)
Diarrhea	23 (15)	15 (10)	4 (3)	4 (3)	12 (8)	7 (5)	3 (2)	2 (1)
Fatigue	19 (13)	7 (5)	11 (7)	1 (1)	27 (17)	20 (13)	5 (3)	2 (1)

The safety population included all randomized patients who received at least one dose of any study treatment.

*****Listed are adverse events that were reported in at least 15% of patients in any group.

Table S8. Incidence and Median Time to First Onset of Treatment-emergent Adverse Events of Special Interest

	Amivantamab-chemotherapy (n=151)				**Chemotherapy** (n=155)			
Event	**All Grades**	**Grade 1**	**Grade 2**	**Grade ≥3**	**All Grades**	**Grade 1**	**Grade 2**	**Grade ≥3**
Infusion-related reaction*								
No. of patients (%)	63 (42)	15 (10)	46 (30)	2 (1)	2 (1)	2 (1)	0	0
No. of hours to first onset — median (range)	0.98 (0.1–8066.5)	1.93 (0.6–4176.7)	0.98 (0.1–8066.5)	7.22 (7.2– 7.2)	5041.05 (5041.1–5041.1)	5041.05 (5041.1–5041.1)	–	–
Rash†								
No. of patients (%)	135 (89)	59 (39)	47 (31)	29 (19)	28 (18)	21 (14)	7 (5)	0
No. of days to first onset — median (range)	10 (1–310)	12 (2–90)	20 (2–423)	105 (14–469)	37.5 (1–341)	25 (1–341)	145 (31–273)	–
Interstitial lung disease‡								
No. of patients (%)	4 (3)	0	0	4 (3)	0	0	0	0
No. of days to first onset — median (range)	128 (13–188)	–	–	128 (13–188)	–	–	–	–

The safety population included all randomized patients who received at least one dose of any study treatment.

*There were no grade 4 or 5 infusion-related reactions.

†Grouping includes the following preferred terms: rash, dermatitis acneiform, dermatitis, rash pustular, acne, folliculitis, rash maculopapular, pustule, rash papular, rash pruritic, skin lesion, erythema, and rash macular; there were no grade 4 or 5 events of rash.

‡Grouping includes the following preferred terms: interstitial lung disease and pneumonitis; there were no grade 4 or 5 events of interstitial lung disease or pneumonitis.

Table S9. Treatment-emergent Serious Adverse Events Occurring in At Least 2 Patients

Event — no. (%)	**Amivantamab-chemotherapy** (n=151)	**Chemotherapy** (n=155)
Pneumonia	6 (4)	4 (3)
Pneumonitis	4 (3)	0
Pulmonary embolism	4 (3)	4 (3)
COVID-19	3 (2)	1 (1)
Hypokalemia	3 (2)	1 (1)
Thrombocytopenia	3 (2)	5 (3)
Vomiting	3 (2)	1 (1)
Asthenia	2 (1)	1 (1)
Cellulitis	2 (1)	1 (1)
Dermatitis acneiform	2 (1)	0
Diarrhea	2 (1)	1 (1)
Neutropenia	2 (1)	0
Rash	2 (1)	0
Rash pustular	2 (1)	0
Skin infection	2 (1)	0
Anemia	1 (1)	6 (4)
Dyspnea	1 (1)	5 (3)
Febrile neutropenia	1 (1)	3 (2)
Pleural effusion	1 (1)	5 (3)

The safety population included all randomized patients who received at least one dose of any study treatment.

Table S10. Treatment-emergent Adverse Events Leading to Treatment Interruptions, Reductions, and Discontinuations

Event — no. (%)	**Amivantamab-chemotherapy** (n=151)	**Chemotherapy** (n=155)
Any event leading to interruptions of any study agent	104 (69)	56 (36)
Most common events leading to interruption of any study agent*		
Neutropenia	25 (17)	11 (7)
Rash	18 (12)	0
Paronychia	13 (9)	0
COVID-19	12 (8)	12 (8)
Hypokalemia	10 (7)	0
Thrombocytopenia	9 (6)	8 (5)
Anemia	8 (5)	11 (7)
Dermatitis acneiform	8 (5)	0
Any event leading to reductions of any study agent	73 (48)	35 (23)
Most common events leading to reductions of any study agent*		
Neutropenia	14 (9)	8 (5)
Rash	14 (9)	0
Paronychia	12 (8)	0
Thrombocytopenia	10 (7)	13 (8)
Any event leading to discontinuations of any study agent	36 (24)	16 (10)
Most common events leading to discontinuations of any study agent†		
Pneumonitis‡	4 (3)	0
Anemia	3 (2)	1 (1)
Decreased appetite	3 (2)	0
Infusion-related reaction	3 (2)	0
Neutropenia	3 (2)	2 (1)
Thrombocytopenia	1 (1)	3 (2)

The safety population included all randomized patients who received at least one dose of any study treatment.

*Listed are adverse events that were reported in at least 5% of patients in any group.

†Listed are adverse events that were reported in at least 3 patients in any group.

‡Per protocol, patients were required to discontinue study treatment for treatment-emergent pneumonitis.

Table S11. Summary and Cause of Death

Event — no. (%)	**Amivantamab-chemotherapy** (n=153)	**Chemotherapy** (n=155)
Deaths during study	28 (18)	42 (27)
Progressive disease	20 (13)	30 (19)
Adverse event	4 (3)	9 (6)
COVID-19	2 (1)	0
Other	2 (1)	3 (2)
Deaths within 30 days of last dose	7 (5)	7 (5)
Progressive disease	1 (1)	2 (1)
Adverse event	3 (2)	4 (3)
COVID-19	2 (1)	0
Other	1 (1)	1 (1)
Deaths within 30 days of last dose (main study)*	7 (5)	4 (3)
Progressive disease	1 (1)	1 (1)
Adverse event	3 (2)	2 (1)
COVID-19	2 (1)	0
Other	1 (1)	1 (1)

The efficacy analysis set included all randomized patients.

*Patients who died during the crossover phase are excluded.

Table S12. All Grade 5 Treatment-emergent Adverse Events*

Event — no. (%)	**Amivantamab-chemotherapy†** (n=151)	**Chemotherapy‡** (n=155)
COVID-19	1 (1)§	0
COVID-19 pneumonia	1 (1)§	0
Cardiorespiratory arrest‖	1 (1)§	0
Cerebrovascular accident	1 (1)§	0
Death¶	1 (1)§	1 (1)
Pneumonia	1 (1)	0
Sepsis	1 (1)§	1 (1)
Acute myocardial infarction	0	1 (1)
Dyspnea	0	1 (1)§

The safety population included all randomized patients who received at least one dose of any study treatment.

*All grade 5 adverse events correspond to treatment-emergent adverse events leading to death and vice versa.

†Three events in the amivantamab-chemotherapy arm were assessed as being related to study treatment; 1 event (cardiorespiratory arrest) was related to amivantamab, and 2 events (COVID-19 pneumonia and sepsis) were related to chemotherapy alone.

‡Two events in the chemo arm were assessed as being related to chemotherapy (1 event of death and 1 event of sepsis).

§Denoting the treatment-emergent adverse events leading to death within 30 days of the last dose that occurred within 90 days after the first dose.

‖Death due to cardiorespiratory arrest occurred in the setting of tumor infiltration of heart and was also deemed related to amivantamab.

¶Reason for death was progressive disease for the 1 patient in amivantamab-chemotherapy arm and unknown for the 1 patient in the chemotherapy arm.

Table S13. Treatment-emergent Adverse Events by Toxicity Grades 1, 2, and 3 or Higher

	Amivantamab-chemotherapy (n=151)				**Chemotherapy (n=155)**			
Event — no. (%)*	All grade	Grade 1	Grade 2	Grade ≥3	All grade	Grade 1	Grade 2	Grade ≥3
Neutropenia	89 (59)	10 (7)	29 (19)	50 (33)	70 (45)	8 (5)	27 (17)	35 (23)
Paronychia	85 (56)	35 (23)	40 (26)	10 (7)	0	0	0	0
Rash	81 (54)	40 (26)	24 (16)	17 (11)	12 (8)	10 (6)	2 (1)	0
Anemia	76 (50)	34 (23)	26 (17)	16 (11)	85 (55)	23 (15)	43 (28)	19 (12)
Infusion-related reactions	63 (42)	15 (10)	46 (30)	2 (1)	2 (1)	2 (1)	0	0
Hypoalbuminemia	62 (41)	9 (6)	47 (31)	6 (4)	15 (10)	11 (7)	4 (3)	0
Constipation	60 (40)	51 (34)	9 (6)	0	47 (30)	34 (22)	12 (8)	1 (1)
Leukopenia	57 (38)	14 (9)	26 (17)	17 (11)	50 (32)	15 (10)	30 (19)	5 (3)
Nausea	55 (36)	41 (27)	13 (9)	1 (1)	65 (42)	48 (31)	17 (11)	0
Thrombocytopenia	55 (36)	26 (17)	14 (9)	15 (10)	46 (30)	16 (10)	14 (9)	16 (10)
Decreased appetite	54 (36)	31 (21)	19 (13)	4 (3)	43 (28)	34 (22)	7 (5)	2 (1)
Alanine aminotransferase increased	50 (33)	35 (23)	9 (6)	6 (4)	56 (36)	38 (25)	16 (10)	2 (1)
Aspartate aminotransferase increased	47 (31)	42 (28)	4 (3)	1 (1)	51 (33)	47 (30)	3 (2)	1 (1)
Dermatitis acneiform	47 (31)	23 (15)	18 (12)	6 (4)	5 (3)	4 (3)	1 (1)	0
Peripheral edema	45 (30)	28 (19)	15 (10)	2 (1)	16 (10)	13 (8)	3 (2)	0
Stomatitis	38 (25)	18 (12)	18 (12)	2 (1)	9 (6)	6 (4)	3 (2)	0
COVID-19	36 (24)	24 (16)	9 (6)	3 (2)	21 (14)	13 (8)	7 (5)	1 (1)
Diarrhea	31 (21)	22 (15)	4 (3)	5 (3)	20 (13)	14 (9)	4 (3)	2 (1)
Hypokalemia	32 (21)	14 (9)	5 (3)	13 (9)	13 (8)	8 (5)	3 (2)	2 (1)
Vomiting	32 (21)	17 (11)	10 (7)	5 (3)	29 (19)	22 (14)	6 (4)	1 (1)
Asthenia	30 (20)	12 (8)	10 (7)	8 (5)	29 (19)	20 (13)	5 (3)	4 (3)
Pyrexia	24 (16)	20 (13)	4 (3)	0	9 (6)	8 (5)	1 (1)	0

Fatigue	23 (15)	8 (5)	14 (9)	1 (1)	32 (21)	25 (16)	5 (3)	2 (1)
Gamma-glutamyltransferase increased	21 (14)	14 (9)	3 (2)	4 (3)	26 (17)	12 (8)	8 (5)	6 (4)
Cough	21 (14)	15 (10)	6 (4)	0	24 (15)	18 (12)	6 (4)	0

The safety population included all randomized patients who received at least one dose of any study treatment.

*Listed are adverse events that were reported in at least 15% of patients in any group.

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